RECIPROCAL AND QUOTIENT RELATIONSHIPS

Write everything in sine and cosine

One of the most reliable first steps when simplifying a trigonometric expression.

BEFORE

Mixed functions

$$\sec\theta \tan\theta$$

Two different functions, neither of them sine or cosine

REWRITE

reciprocal for secant, quotient for tangent

SHOW WORKING

AFTER

Sine and cosine only

$$\frac{\sin\theta}{\cos^2\theta}$$

One fraction in the two base functions

WHAT CHANGED

Each function was replaced by its form in sine and cosine, and the product collapsed into one

WHAT STAYED

The value, at every angle where both secant and tangent are defined.

fraction.